

HoloPaws: Run - Diegetic and Non-Diegetic User Interfaces for Mixed Reality Fitness Gaming

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Abstract

Using Microsoft's *HoloLens 2*, this work aims to explore how diegetic and non-diegetic elements impact player motivation, engagement, and performance in outdoor running fitness games. We developed three setups: an *Android* fitness app and two versions of a *HoloLens* game: diegetic and non-diegetic. The *Android* app provides real-time step data and cadence feedback. The non-diegetic *HoloLens* game displays feedback on the user interface via a cadence bar, step counter, and pop-up messages. In contrast, the diegetic *HoloLens* game uses a virtual dog companion to communicate feedback by running alongside the player or sitting when cadence drops, fully integrated into the outdoor running experience. In a future publication we will detail a mixed-methods user study of these three interfaces. We aim to learn insights into how different feedback styles affect performance, motivation, and user experience during outdoor running activities. We hypothesize that diegetic elements can enhance engagement in mixed-reality fitness applications.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Interaction design**; **Interaction techniques**; • **Applied computing** → **Health informatics**.

Keywords

Mixed Reality (MR), Augmented Reality (AR), Head-Mounted Displays (HMDs), Gamification, Fitness Tracking Technologies, Adaptive User Interfaces, Real-time Motion Feedback, Multisensory Feedback

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1 Introduction

The global crisis of physical inactivity presents an urgent challenge, with approximately 25% of adults and 80% of adolescents worldwide leading predominantly sedentary lifestyles [5]. Despite established evidence that regular physical activity enhances health outcomes [8, 23], traditional exercise methods often fail to engage and motivate users effectively. While various technological solutions have emerged to address this challenge, most current approaches lack the immersive and interactive elements needed to sustain long-term engagement. In response to these limitations, research needs to be done to explore the potential of Mixed Reality (MR) technology to transform physical activity into an engaging, gamified experience.

MR creates environments where physical and digital objects coexist and interact in real time, offering unique advantages over both Virtual Reality (VR), which immerses users in fully artificial environments, and Augmented Reality (AR), which merely overlays digital information onto the real world [24]. By leveraging these capabilities, our research aims to investigate how immersive technologies can enhance exercise motivation and adherence through interactive, adaptive feedback mechanisms.

Recent technological advances in Extended Reality (XR) systems have enabled sophisticated real-time feedback mechanisms and immersive learning platforms, incorporating visual, tactile, and haptic feedback for motion optimization and performance correction [16]. While existing fitness applications and wearable devices provide valuable health metrics tracking, they often lack immersive, real-time engagement features. Current solutions like *Fitbit*, *Garmin*, and *Apple Watch* offer continuous health monitoring but face limitations in providing adaptive feedback [1–3]. Similarly, popular fitness apps such as *MyFitnessPal* and *Strava*, while useful for workout tracking, generally fail to deliver truly interactive experiences that maintain user engagement [4, 6].

This work presents *HoloPaws: Run*, a MR fitness application that transforms traditional exercise tracking through three distinct implementations: *PawTrack*, a smartphone-based solution; *Cadence Quest*, featuring non-diegetic *HoloLens* overlays; and *Run Rover*, incorporating an interactive virtual companion. By comparing diegetic and non-diegetic elements in fitness applications, this research aims to demonstrate the evolution from conventional tracking to immersive MR experiences [25]. The key contributions of this work are:

(1) Adaptive Framework Development

- Real-time performance-based workout intensity adjustment
- Integrated step count and cadence tracking system

- Dynamic difficulty scaling mechanisms
- (2) **Novel Feedback Mechanisms**
- Implementation of a virtual dog companion as a diegetic feedback system
 - Integration of performance metrics into the narrative environment
 - Comparison of diegetic versus non-diegetic User Interface (UI) approaches
- (3) **Progressive Technology Integration**
- Evolution from traditional tracking to immersive MR experiences
 - Framework for future adaptive fitness applications
 - Enhanced understanding of MR's role in physical activity engagement

2 Related Works

While VR platforms like Oculus *Quest* have dominated immersive fitness gaming [9], research in MR applications for physical activity remains underexplored. Previous studies primarily utilized smartphone-based or projection-based AR solutions, with findings suggesting that MR's blend of virtual and physical elements could enhance exercise engagement while maintaining environmental awareness [12]. Our research leverages the Microsoft *HoloLens 2*'s advanced capabilities to address limitations in natural interaction and real-world integration identified in previous studies.

2.1 Exergaming and Feedback Mechanisms

Exercise in static environments, particularly on treadmills, often suffers from declining engagement due to limited environmental stimuli [19]. Research by Graf et al. [15] investigated how emotional virtual coaches affect user motivation and performance, finding that both happy and angry virtual coaches significantly influenced player emotions and heightened motivation. Similarly, Fu et al. [13] demonstrated the effectiveness of VR games like *VRrowing* in promoting physical and cognitive health for specific demographics. Kaplan et al. [16] explored age-specific responses to different difficulty adjustment methods, revealing that older individuals responded better to ramping and biofeedback adjustments, while younger users benefited more from performance-based modifications. This research highlights the importance of age-appropriate design considerations in exergames.

2.2 Interactive and Adaptive Design

Recent studies have emphasized the critical role of user satisfaction in AR applications. H. Xue et al. [28], Sheehan et al. [22], and Zhou et al. [29] demonstrated its significance in interactive training scenarios. Our research extends these findings by investigating satisfaction through both device interaction and virtual companion engagement. Pedrolí et al. [21] developed *Positive Bike*, integrating a cycle-ergometer with VR for elderly rehabilitation. Their findings on virtual environment interaction and proactive feedback inform our approach to user engagement. Additionally, Castillo et al. [26] employed user-centered design in developing VR rehabilitation games, providing valuable insights for adaptive feedback systems.

2.3 Real-time Feedback in MR

Geisen et al. [14] outlined the advantages of real-time XR feedback in sports training, proposing a framework for integrating immediate corrections into motor learning. This aligns with our implementation of real-time feedback mechanisms in fitness gaming. Klasnja et al. [17] investigated how physical activity and gaming affect reaction time, finding significant differences between activity types.

2.4 VR Fitness Applications

Recent developments in VR fitness applications provide valuable insights for our MR approach. Craig et al. [11] evaluated energy expenditure in *Supernatural*'s Flow and Boxing modes, confirming their qualification as vigorous physical activities. Xu et al. [27] developed *VirusBoxing*, combining High-Intensity Interval Training (HIIT) with immersive gameplay, while Cioffi et al. [10] examined long-term effects of VR exercise on anxiety and executive function.

2.5 Diegetic and Non-Diegetic Elements

Beck et al. [7] demonstrated how diegetic cues enhance User Experience (UX) in interactive VR environments. Kohle et al. [18] compared different health interface implementations, finding that context-specific design significantly impacts user experience. These studies inform our approach to integrating feedback mechanisms within the game world. Table 1 provides a comprehensive comparison of existing fitness applications with *HoloPaws: Run*, highlighting our unique contributions to the field. Our research builds upon these studies by combining MR technology's strengths with interactive design principles, creating fitness experiences that seamlessly integrate real-time feedback, adaptive mechanics, and engaging diegetic elements to promote physical activity.

2.6 Comparison with Related Work

Table 1 summarizes key fitness games and applications, comparing their features with *HoloPaws: Run*. Unlike VR games that isolate users, our MR solution maintains environmental awareness and supports outdoor activities, addressing limitations in traditional fitness applications.

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3 HoloPaws: Run

3.1 PawTrack: An Android Application for Fitness Tracking and HoloLens Integration

PawTrack is an *Android* app developed for real-time fitness tracking and integration with MR experiences on *HoloLens*. Using the Google Fit API, it captures step counts and cadence, sending this data to the *HoloLens* for interactive gameplay. The app explores user preferences for fitness interfaces, comparing the ease of smartphone use with the immersive, hands-free capabilities of *HoloLens*.

The app's name reflects its integration with a virtual dog companion in both platforms, engaging users through gamification. *PawTrack* will serve as the control condition in our study which

Table 1: Comparison of fitness games and applications with *HoloPaws: Run*

App/Game	Platform	Features	Comparison to <i>HoloPaws</i>
<i>Supernatural</i> [11]	VR	Vigorous activity	Indoor-focused; lacks MR outdoor integration.
<i>VRrowing</i> [13]	VR	Rowing for seniors	Focus on older adults; <i>HoloPaws</i> targets Emerging Adults (EAs).
Marin-Lora et al. [20]	Exergame	Treadmill running	Emphasizes indoor use; <i>HoloPaws</i> supports outdoor fitness.
Graf et al. [15]	VR	Emotional coaching	Emotional feedback informs virtual companion design.
Craig et al. [11]	VR	Mood improvements	<i>HoloPaws</i> incorporates outdoor interaction and real-world integration.

will be detailed in a future publication, providing a baseline for comparing immersive and conventional fitness tracking methods.

3.1.1 Architecture and Design. *PawTrack* employs a client-server model. The client-side, developed in *Kotlin*, interfaces with fitness sensors, collecting metrics like step count and cadence. User inputs such as age, height, and step goals further personalize the experience. The server-side, implemented in *C#*, processes the transmitted data on *HoloLens*, dynamically influencing gameplay and virtual companion behavior.

3.1.2 Key Features and Data Integration. Real-time metrics like step count and cadence drive user engagement. For instance, cadence data adjusts the speed of the virtual dog, creating an adaptive experience (Figure 1). The focus was mainly on step count and cadence for real-time responsiveness. Collected data is transmitted to *HoloLens* via socket communication, ensuring minimal latency and seamless synchronization. This setup enhances the user's fitness journey by integrating real-world data into an immersive MR environment.

3.2 *HoloPaws: Cadence Quest* - Gamified UI Feedback with Virtual Dog Companion

HoloPaws: Cadence Quest represents our initial approach to MR fitness gaming, combining real-time fitness tracking with gamification elements. The game guides players through a structured progression of three distinct fitness levels: brisk walking, jogging, and running, each calibrated to specific cadence thresholds. This tiered approach ensures a gradual increase in physical activity intensity while maintaining user engagement.

The game's interface integrates several dynamic UI elements that respond in real time to the player's physical activity which are shown in Figure 2:

- A responsive cadence bar functions as a visual speedometer, dynamically expanding or contracting based on the user's current pace
- Pop-up notifications mark significant transitions between activity levels
- Real-time metrics display, including step count, calories burned, and current activity level
- A virtual dog companion that synchronizes its movement with the player's pace

3.2.1 Technical Implementation. The game receives real-time fitness data through the *Android* app's integration with Google Fit API. This data stream provides continuous updates on step count and cadence, which the game transforms into visual feedback elements. The implementation focuses on minimal latency to ensure responsive feedback, with the game processing and displaying updates immediately as new data arrives from the mobile application.

3.2.2 Game Progression. Players begin by setting a personal step goal, which the game automatically divides into three phases: approximately 33% for brisk walking, 34% for jogging, and 33% for running. This distribution ensures balanced progression through different intensity levels. The virtual dog companion runs ahead of the player, maintaining synchronization with their movement speed, while the UI provides numerical feedback through the cadence bar and step counter.

The game tracks calories burned using player demographics (age, height, weight) and activity intensity, displaying this information periodically on a separate canvas to avoid cluttering the main interface. Achievement notifications appear at key milestones, celebrating the player's progress through each activity phase.

3.3 *HoloPaws: Run Rover* - Diegetic Feedback Through Dynamic Dog Behavior

Run Rover represents an evolution in our approach to fitness feedback, shifting from traditional UI elements to a fully diegetic system where the virtual dog serves as the primary feedback mechanism. This design choice emphasizes natural, intuitive feedback integrated seamlessly into the game environment.

3.3.1 Advanced Behavioral Mechanics. The dog's behavior system responds to specific cadence thresholds with carefully designed actions:

- Below target cadence: The dog runs ahead, sits, and waits for the player
- Optimal cadence range: The dog runs alongside the player, maintaining pace
- Above target cadence: The dog moves to the player's left and barks as a signal

The virtual companion's actions include synchronized audio feedback, with panting sounds that adjust to match the player's

pace, creating a more immersive experience. This multi-modal feedback approach (visual, spatial, and auditory) helps users naturally understand and respond to their performance level.

3.3.2 Design Philosophy. While both games track identical fitness metrics, their presentation differs significantly. *Cadence Quest* employs traditional UI elements for precise numerical feedback, while *Run Rover* embeds this information within the dog’s behavioral patterns. This contrast allows us to evaluate whether natural, embodied feedback can achieve similar or superior engagement compared to conventional interface designs.

The elimination of traditional UI elements in *Run Rover* represents a deliberate design choice to test the effectiveness of pure diegetic feedback in maintaining user motivation and understanding of performance metrics. This approach aligns with research suggesting that naturalistic feedback can enhance immersion and emotional engagement in fitness applications.

4 Preliminary Study

A preliminary pilot study with five participants evaluated two versions of *HoloPaws: Run Rover* to inform design decisions and optimize the virtual companion’s behavior patterns. The study focused specifically on implementing the dog’s behavior as an effective diegetic element within the game environment.

Since the testing took place outdoors, one challenge encountered was that the *HoloLens 2* would not perform optimally in direct sunlight. To mitigate this issue, we experimented with three different shades of Neutral Density (ND) filters—ND3, ND6, and ND9—to improve visibility. After testing, the best results were achieved using the ND6 and ND9 filters, which provided the right balance of light reduction without compromising display clarity. These filters were cut to size and applied directly to the *HoloLens 2*, allowing participants to use the device effectively under bright sunlight conditions.

4.1 Participants

Five participants (two females, three males), aged 27 - 38 were recruited from a graduate computer science student population. The weight range of the participants was 47 kg to 96 kg and their heights, ranged from 153 to 192 cm. Four participants were new to *HoloLens 2*, while one had prior knowledge. Two participants had pets and had high fitness levels, and the others had moderate fitness level.

4.2 Methodology

Participants tested two game versions: one featuring basic dog animations for a relaxed experience, and another incorporating enhanced interaction with synchronized movements and audio feedback. The study collected participant feedback through direct observation and semi-structured interviews.

4.3 Key Findings

Three participants found value in both versions, noting distinct use cases for each implementation. The basic version provided a "simple and relaxing" experience, while the enhanced version offered greater engagement through synchronized audio-visual feedback. This synchronization proved particularly effective in

maintaining participant running pace. Participants emphasized the importance of maintaining numerical data displays alongside the dog’s movements, though two users reported occasional cognitive overload from multiple simultaneous interface elements in the basic version. Usage context emerged as a significant factor, with participants indicating their version preference would vary based on intended workout intensity.

4.4 Design Implications

The pilot study findings directly influenced the final design decisions for *HoloPaws: Run Rover*, particularly in:

- Implementing synchronized audio-visual feedback while carefully managing cognitive load
- Balancing numerical data display with diegetic elements
- Maintaining flexibility in feedback intensity to accommodate different exercise intentions

These insights suggested that preserving both interaction modes could effectively serve varying user needs and exercise goals.

Cognitive Load Considerations. One of the key trade-offs between diegetic and non-diegetic feedback mechanisms is cognitive load. Traditional UI elements explicitly present information, but they require the user to process multiple visual indicators simultaneously. In contrast, diegetic feedback—such as the dog’s movements—relies on more intuitive, natural responses. However, our preliminary study indicated that some participants found it difficult to interpret the dog’s behavior under intense exercise conditions, while others experienced cognitive overload when too many UI elements were present. Future iterations should explore ways to balance numerical data presentation with embedded environmental cues, ensuring that feedback remains accessible yet minimally intrusive.

4.5 Practical Considerations for Outdoor Use

MR exergames face challenges in outdoor environments. Pass-through VR headsets, such as the *Meta Quest Pro* and *HTC Vive XR Elite*, use external cameras to display the real world, but their resolution and clarity remain inferior to dedicated MR headsets. These systems often suffer from low resolution, distortion, and narrow Field of View (FOV), reducing situational awareness. Motion blur and latency can also cause disorientation, making navigation difficult and increasing accident risks.

The *HoloLens 2*, in contrast, uses a transparent display, allowing users to see their environment while interacting with digital elements. Its high-resolution optics improve real-world interaction, making it a safer option for outdoor fitness applications. The *HoloLens 2* also employs true spatial mapping to create a 3D model of the environment, enhancing depth perception and obstacle detection.

However, outdoor use presents challenges. Direct sunlight reduces display visibility, but ND filters (ND3, ND6, ND9) improve clarity while maintaining transparency. ND6 and ND9 provided the best balance between ambient light reduction and display visibility.

User safety remains a concern. While MR maintains environmental awareness, prolonged digital engagement can still be distracting. Unlike traditional fitness apps that allow brief glances, MR interfaces require sustained focus. Pass-through VR further

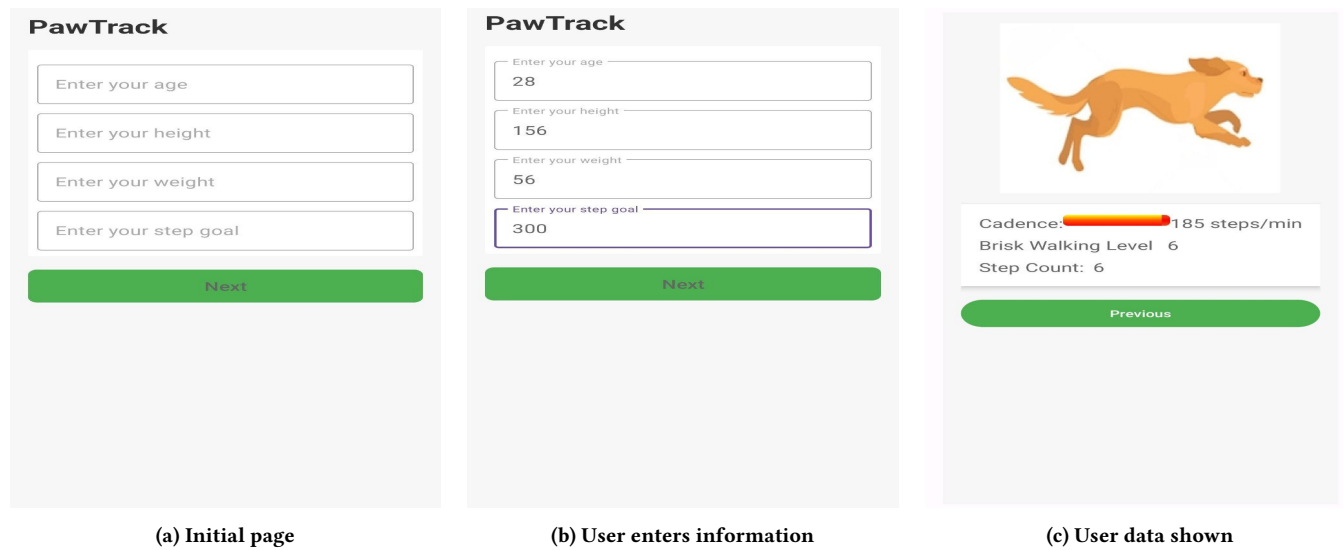
Figure 1: The UI of *PawTrack*

Figure 2: Goals reached for various activities

limits awareness by replacing the real world, whereas the *HoloLens 2* preserves visibility. Still, users may experience disorientation, particularly in fast-moving activities.

To enhance outdoor exergames, future implementations should consider:

- Adaptive UI elements that adjust brightness for visibility in varying light conditions.

- Audio or haptic feedback to warn users of nearby hazards.
- Peripheral UI placement to minimize distraction.
- Integration of spatial mapping for real-time obstacle detection.

5 Conclusion

This research has demonstrated the potential of MR in transforming fitness experiences through the development and evaluation of *HoloPaws: Run*. Through our preliminary study, we established the effectiveness of diegetic elements in enhancing user engagement, particularly through the virtual companion's synchronized audio-visual feedback. The progression from traditional smartphone tracking to immersive MR experiences revealed valuable insights into user preferences and interaction patterns.

The findings of our preliminary study highlight several key contributions to the field of MR fitness applications. First, the implementation of a virtual companion as a primary feedback mechanism represents a novel approach to fitness motivation, naturally integrating performance metrics into the exercise experience. Second, the comparison between diegetic and non-diegetic elements provided crucial insights into cognitive load management and user engagement. Third, the development of three distinct implementations—*PawTrack*, *Cadence Quest*, and *Run Rover*—established a comprehensive framework for evaluating user engagement across different interaction modalities.

However, the study had several limitations, including the small participant sample, outdoor visibility constraints of the *HoloLens 2*, and potential safety concerns for outdoor use. Addressing these challenges will be crucial for future research to improve real-world adoption and scalability of MR fitness applications.

6 Future Work

Several key directions remain for future work. First, a larger-scale user study with a more diverse participant pool is necessary to validate our preliminary findings. This study should include a structured evaluation using standardized metrics such as User Experience Questionnaire (UEQ) and NASA Task Load Index (NASA-TLX) to assess usability, engagement, and cognitive load more rigorously.

Second, safety considerations for outdoor use must be further explored. Although ND filters improved visibility in bright environments, additional solutions, such as adaptive display brightness or peripheral safety notifications, could improve usability. Furthermore, implementing environmental awareness mechanisms, such as obstacle detection or audio warnings, could help mitigate outdoor safety risks.

Third, long-term engagement patterns should be investigated through longitudinal studies. Understanding whether users maintain motivation over extended periods and whether diegetic feedback remains effective or fades due to novelty effects will be critical to refining the system. Furthermore, incorporating adaptive difficulty levels based on real-time performance and physiological data (e.g., heart rate variability) could improve personalized training experiences.

Finally, social and multiplayer aspects of *HoloPaws: Run* present an exciting avenue for future research. Introducing shared virtual

companions, collaborative running modes, or competitive challenges could further increase engagement. However, it is essential to balance immersion with safety and usability concerns in multi-user scenarios.

By addressing these areas, we aim to refine *HoloPaws: Run* into a fully adaptive MR fitness experience ready for the real world that enhances user motivation and engagement.

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References

- [1] [n. d.]. Apple Watch. <https://www.apple.com/watch/>. Accessed: 2024-10-31.
- [2] [n. d.]. Garmin - Wearables and Smartwatches. <https://www.garmin.com/en-US/c/wearables-smartwatches/>. Accessed: 2024-10-31.
- [3] [n. d.]. Google Store - Watches and Trackers. https://store.google.com/category/watches_trackers?hl=nl&pli=1. Accessed: 2024-10-31.
- [4] [n. d.]. MyFitnessPal. <https://www.myfitnesspal.com/>. Accessed: 2024-10-31.
- [5] [n. d.]. Physical activity — who.int. <https://www.who.int/news-room/fact-sheets/detail/physical-activity>. [Accessed 27-09-2024].
- [6] [n. d.]. Strava. <https://www.strava.com/>. Accessed: 2024-10-31.
- [7] Thomas Beck and Sylvia Rothe. 2021. Applying diegetic cues to an interactive virtual reality experience. In *2021 IEEE Conference on Games (CoG)*, 1–8. doi:10.1109/CoG52621.2021.9619025
- [8] Claude Bouchard and Tuomo Rankinen. 2001. Individual differences in response to regular physical activity. *Medicine & Science in Sports & Exercise* 33, 6 (2001), S446–S451.
- [9] Luis Emilio Bruni, Nele Kadastik, Thomas Anthony Pedersen, and Hossein Dini. 2022. Digital Narratives in Extended Realities. *Roadmapping Extended Reality: Fundamentals and Applications* (2022), 35–62.
- [10] Rose Cioffi and Anat V Lubetzky. 2023. BOXVR versus guided YouTube boxing for stress, anxiety, and cognitive performance in adolescents: A pilot randomized controlled trial. *Games for Health Journal* 12, 3 (2023), 259–268. doi:10.1089/g4h.2022.0202
- [11] Tabitha V Craig, Ryan E Rhodes, and Wuyou Sui. 2024. Examining and Comparing the Energy Expenditure of Two Modes of a Virtual Reality Fitness Game (Supernatural): Indirect Calorimetry Study. *JMIR Serious Games* 12 (4 Jun 2024), e53999. doi:10.2196/53999
- [12] Opeyemi Dele-Ajayi, Jonathan Sanderson, Rebecca Strachan, and Alison Pickard. 2016. Learning mathematics through serious games: An engagement framework. In *2016 IEEE Frontiers in Education Conference (FIE)*, 1–5. doi:10.1109/FIE.2016.7757401
- [13] Yu Fu, Yan Hu, Veronica Sundstedt, and Yvonne Forsell. 2022. Conceptual Design of an Extended Reality Exercise Game for the Elderly. *Applied Sciences* 12, 13 (2022). doi:10.3390/app12136436
- [14] Mai Geisen and Stefanie Klatt. 2022. Real-time feedback using extended reality: A current overview and further integration into sports. *International Journal of Sports Science & Coaching* 17, 5 (2022), 1178–1194. doi:10.1177/17479541211051006 arXiv:https://doi.org/10.1177/17479541211051006
- [15] Linda Graf, Sophie Abramowski, Felix Born, and Maic Masuch. 2023. Emotional Virtual Characters for Improving Motivation and Performance in VR Exergames. *Proc. ACM Hum.-Comput. Interact.* 7, CHI PLAY, Article 417 (oct 2023), 21 pages. doi:10.1145/3611063
- [16] Oral Kaplan, Goshiro Yamamoto, Takafumi Taketomi, Alexander Plopski, Christian Sandor, and Hirokazu Kato. 2018. Exergame Experience of Young and Old Individuals Under Different Difficulty Adjustment Methods. *Computers* 7, 4 (2018). doi:10.3390/computers7040059
- [17] Aleksandar Klasnja, Natasa Milenovic, Sonja Lukac, Aleksandar Knezevic, Jelena Klasnja, and Vedrana Karan Rakic. 2022. The Effects of Regular Physical Activity and Playing Video Games on Reaction Time in Adolescents. *International Journal of Environmental Research and Public Health* 19, 15 (July 2022), 9278. doi:10.3390/ijerph19159278
- [18] Kay Köhle, Matthias Hoppe, Albrecht Schmidt, and Ville Mäkelä. 2021. Diegetic and Non-diegetic Health Interfaces in VR Shooter Games. In *Human-Computer Interaction – INTERACT 2021*, Carmelo Ardito, Rosa Lanzilotti, Alessio Malizia, Helen Petrie, Antonio Piccinno, Giuseppe Desolda, and Kori Inkpen (Eds.). Springer International Publishing, Cham, 3–11.

- [19] Namsu Lee, Wonjae Choi, and Seungwon Lee. 2021. Development of an 360-degree virtual reality video-based immersive cycle training system for physical enhancement in older adults: a feasibility study. *BMC Geriatrics* 21 (05 2021). doi:10.1186/s12877-021-02263-1
- [20] Carlos Marín-Lora, Miguel Chover, Micaela Yanet Martín, and Linda García-Rytman. 2023. Creating a treadmill running video game with smartwatch interaction. *Multimedia Tools and Applications* 83, 19 (Dec. 2023), 57709–57729. doi:10.1007/s11042-023-17752-1
- [21] Elisa Pedrolí, Luca Greci, Desirée Colombo, Silvia Serino, Pietro Cipresso, Sara Arlati, Marta Mondellini, Lorenzo Boilini, Valentina Giussani, Karine Goulene, Monica Agostoni, Marco Sacco, Marco Stramba-Badiale, Giuseppe Riva, and Andrea Gaggioli. 2018. Characteristics, Usability, and Users Experience of a System Combining Cognitive and Physical Therapy in a Virtual Environment: Positive Bike. *Sensors* 18, 7 (2018). doi:10.3390/s18072343
- [22] Dwayne Sheehan and Larry Katz. 2010. Using interactive fitness and exergames to develop physical literacy. *Physical & Health Education Journal* 76, 1 (2010).
- [23] Barbara J Speck and Joanne S Harrell. 2003. Maintaining regular physical activity in women: evidence to date. *Journal of Cardiovascular Nursing* 18, 4 (2003), 282–293.
- [24] Maximilian Speicher, Brian D. Hall, and Michael Nebeling. 2019. What is Mixed Reality?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3290605.3300767
- [25] Taiyo Taguchi and Tomokazu Ishikawa. 2022. A Research On The Compatibility Between MR/VR Devices And Exergaming. In *2022 Nicograph International (NicoInt)*. 17–20. doi:10.1109/NicoInt55861.2022.00010
- [26] Julian Felipe Villada Castillo, Maria Fernanda Montoya Vega, John Edison Muñoz Cardona, David Lopez, Leonardo Quiñones, Oscar Alberto Henao Gallo, and Jose Fernando Lopez. 2024. Design of Virtual Reality Exergames for Upper Limb Stroke Rehabilitation Following Iterative Design Methods: Usability Study. *JMIR Serious Games* 12 (11 Jan 2024), e48900. doi:10.2196/48900
- [27] Wenge Xu, Hai-Ning Liang, Xiaoyue Ma, and Xiang Li. 2020. VirusBoxing: A HIIT-based VR Boxing Game. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play* (Virtual Event, Canada) (CHI PLAY '20). Association for Computing Machinery, New York, NY, USA, 98–102. doi:10.1145/3383668.3419958
- [28] Helene Xue, Puneet Sharma, and Fridolin Wild. 2018. User Satisfaction in Augmented Reality-Based Training Using Microsoft HoloLens. doi:10.20944/preprints201810.0594.v2 Preprint.
- [29] Jun-Ye Zhou and Tz-Heng Fu. 2023. Empowering Interactive Fitness Game. In *2023 International Conference on Consumer Electronics-Taiwan (ICCE-Taiwan)*. IEEE, 279–280.